Conditioning-Induced Value Overestimation in Variable-Rate World-Model Imagination

Anonymous Authors

Abstract

Temporal abstraction in world-model reinforcement learning promises improved planning efficiency by allowing agents to imagine at variable time scales. We present a systematic diagnostic analysis of variable-rate imagination in DreamerV3, revealing a failure mode we term *Conditioning-Induced Value Overestimation* (CIVO). Through a factorial experiment spanning 31 configurations across four environments (Walker Walk, Cheetah Run, Cartpole Swingup, Hopper Hop), we establish a precise causal chain: a *trained* temporal embedding (`dt_emb`) enables the critic to exploit imprecise multi-step predictions from the RSSM world model, triggering a positive feedback loop of value overestimation and policy degradation. We introduce Normalized Value Excess (NVE) as a diagnostic metric and show that an NVE threshold of 1.5× achieves AUROC=0.969 in separating pathological from healthy training runs across 18 seeds. Crucially, all attempted interventions fail to robustly eliminate CIVO: gradient-based interventions (selective gradient stopping, 1/17 seeds protected) operate at the wrong abstraction level, while forward-pass interventions (temporal conditioning noise, uncertainty-gated truncation) exhibit narrow effective windows and seed-dependent outcomes ($\leq$1/8 cross-seed-environment prevention). The contrast with TAWM—where dt-conditioning succeeds in MPC-based planning—localizes the vulnerability to the imagination-based actor-critic architecture: critic bootstrapping from imagined trajectories creates the positive feedback loop that MPC's re-planning from real states avoids. Our results establish that robust variable-rate imagination requires architectural changes beyond surface-level fixes, pointing toward hierarchical temporal models as the viable path forward.

1 Introduction

Temporal abstraction—the ability to reason and act at multiple time scales—is a foundational challenge in reinforcement learning. The options framework [Sutton et al., 1999], FiGAR [Sharma et al., 2017], and action repeat mechanisms have demonstrated its value in model-free settings. World-model RL, exemplified by the Dreamer family [Hafner et al., 2020, 2021, 2023], offers a particularly natural testbed: by modifying the temporal granularity of *imagined* rollouts without changing environment interaction, one can decouple the planning horizon from the data collection rate.

Despite this appeal, our systematic attempts to introduce variable-rate imagination into DreamerV3 consistently degrade performance. Fixed $\Delta t = 2$ imagination collapses from a peak of 485 to 135 eval return; fixed $\Delta t = 4$ achieves only 33% of baseline; learned temporal gating peaks at 541 before collapsing to 99. No configuration involving $\Delta t > 1$ imagination matches the $\Delta t = 1$ baseline ($\sim$950 eval return).

This paper asks *why*. Through a factorial experiment design spanning 31 configurations and four DeepMind Control Suite environments, we trace the failure to a specific mechanism we term *Conditioning-Induced Value Overestimation* (CIVO). The causal chain is:

1. A *trained* temporal embedding (`dt_emb`) learns to precisely distinguish $\Delta t = 1$ from $\Delta t > 1$ transitions in the RSSM latent space.
2. RSSM prediction quality degrades at $\Delta t > 1$ (reward prediction gap: 2.78× for trained `dt_emb` vs. 1.56× baseline), but the trained embedding enables the critic to condition on Δt with high precision.
3. The combination of precise conditioning and imprecise predictions creates systematic value overestimation: the critic assigns inflated values to states reached via $\Delta t > 1$ imagination steps.
4. This overestimation drives the actor toward policies that exploit the overestimated regions, creating a positive feedback loop that culminates in performance collapse.

We introduce *Normalized Value Excess* (NVE)—the ratio of a configuration's value-to-eval ratio to the baseline's at the same training step—as a diagnostic metric. An NVE threshold of 1.5× achieves AUROC=0.969 across 18 seeds, reliably identifying pathological overestimation before collapse onset.

A critical contrast motivates our analysis. TAWM [Nhu et al., 2025] successfully uses Δt-conditioning in TD-MPC2, an MPC-based world-model method. The divergence is architectural: MPC re-plans from *real* states at each decision point, breaking the feedback loop; imagination-based actor-critic methods bootstrap critic values from imagined trajectories, sustaining it. This localizes CIVO as an architecture-specific vulnerability, not a fundamental limitation of temporal conditioning.

Our contributions are:

1. A causal diagnosis of CIVO through controlled factorial experiments, establishing trained `dt_emb` as the necessary and sufficient trigger (C1–C5).
2. Discovery of dual vulnerability pathways: classic CIVO (value-driven, entropy decreasing) and non-CIVO policy instability (eval-driven, entropy increasing), demonstrating that the trained embedding creates fragility beyond the overestimation loop (C6).
3. Evidence that gradient-level interventions are categorically insufficient (1/17 seeds across 3 SGS variants), establishing forward-pass representational damage as the irreducible vulnerability floor (C7).
4. A comprehensive negative-result catalog of intervention attempts (12 strategies, $\leq 1/8$ combined prevention rate), providing actionable guidance for future work.

2 Background and Related Work

Temporal abstraction in RL. The options framework [Sutton et al., 1999] formalized hierarchical temporal abstraction, inspiring methods from option-critic [Bacon et al., 2017] to discovery-based approaches. FiGAR [Sharma et al., 2017] and dynamic action repetition learn per-step skip rates in model-free settings. These approaches modify *action execution* rather than *model imagination*.

World-model RL. DreamerV3 [Hafner et al., 2023] learns a Recurrent State-Space Model (RSSM) and trains actor-critic networks entirely within imagined trajectories. IRIS [Micheli et al., 2023] and TWM [Robine et al., 2023] use transformer-based world models. Variable-rate planning has been explored in THICK [Kim et al., 2024] and Hieros [Röpke et al., 2024], which use hierarchical dynamics models with separate temporal scales. TAWM [Nhu et al., 2025] demonstrates successful Δt-conditioned planning in TD-MPC2's MPC framework.

Value overestimation in MBRL. Model exploitation and value overestimation are known challenges in model-based RL [Janner et al., 2019]. Delusional bias [Lu et al., 2018] refers to value overestimation from policy-class constraints—a distinct mechanism from our CIVO, which arises from model imprecision amplified by learned temporal conditioning. Ensemble disagreement and pessimistic value estimation have been proposed as mitigations for general model exploitation, but our results show that CIVO's feedback loop structure resists standard remedies.

3 Methodology

3.1 Base Architecture

We build on DreamerV3 [Hafner et al., 2023] with its RSSM world model, symlog-transformed critic, and actor-critic training within imagined rollouts. The standard configuration uses $\Delta t = 1$ for all imagination steps with horizon $H = 15$.

3.2 Variable-Rate Imagination

We extend the imagination procedure to support $\Delta t > 1$ steps via three mechanisms:

- **Temporal embedding** (`dt_emb`): A learned embedding vector $e(\Delta t) \in \mathbb{R}^d$ concatenated to the RSSM input at each imagination step, conditioning state transitions on the temporal skip.

- **Temporal subsampling**: Training data construction that includes $\Delta t > 1$ transitions by skipping intermediate frames, exposing the RSSM to multi-step prediction during training.
- **Learned gating**: A policy head that selects $\Delta t \in \{1, 2\}$ at each imagination step.

3.3 Factorial Experiment Design

Our core diagnostic uses a factorial grid over three binary factors:

Table 1: Factorial design: core configurations. Each cell denotes a distinct experiment.

		`dt_emb` Training	
Subsample	Horizon	Trained	Untrained/Absent
ON	$H = 8$	008d (collapse)	008f (stable, degraded)
ON	$H = 15$	006 (collapse)	—
OFF	$H = 8$	—	008c (stable, degraded)
OFF	$H = 15$	—	008a (stable, 85% baseline)

Beyond the factorial core, we test additional configurations including frozen `dt_emb`, $\Delta t = 1$ imagination with trained `dt_emb` (008e_v3), and cross-environment replication on Cheetah Run, Cartpole Swingup, and Hopper Hop. All experiments use 3–5 seeds unless noted.

3.4 Diagnostic Metrics

Normalized Value Excess (NVE). We define $\text{NVE}(t) = \frac{r_{\text{ve}}^{\text{exp}}(t)}{r_{\text{ve}}^{\text{baseline}}(t)}$, where $r_{\text{ve}}(t) = V(s_t)/R_{\text{eval}}(t)$ is the value-to-eval ratio. NVE normalizes for the natural increase in r_{ve} during healthy training. $NVE > 1.5\times$ at training step $\geq$ 35K (after baseline r_{ve} stabilizes) serves as our diagnostic threshold for pathological overestimation.

4 Experiments

4.1 C1: Variable-Rate Imagination Consistently Degrades Performance

All $\Delta t > 1$ configurations degrade asymptotic performance relative to the $\Delta t = 1$ baseline across all tested environments.

Table 2: C1: Performance degradation across configurations and environments. Baseline eval returns: Walker ∼950, Cheetah ∼594, Cartpole ∼852.

Configuration	Walker Eval	Cheetah Eval	Cartpole Eval	% Baseline
Baseline (Δt=1)	950	594	852	100%
Fixed Δt=2, trained `dt_emb` (006)	135 (collapse)	—	—	14%
Fixed Δt=4 (007)	144	—	—	15%
Learned gate (005)	99 (collapse)	—	—	10%
Trained `dt_emb` + subsample (008d)	83–268	0.8–52	55–485	9–51%

Cheetah Run baseline. Five seeds converge to a mean of 593.9 (range: 520.3–645.4), with value-to-eval ratio stabilizing at 0.5–0.7, consistent with Walker. This establishes the cross-environment reference.

Cartpole Swingup baseline. Three seeds converge to a mean of 852 (range: 838–865), with faster learning than locomotion tasks (∼80% performance at 35K vs. 65K for Walker).

Table 3: Cheetah Run baseline across seeds.

Seed	s0	s1	s2	s3	s4
Eval (final)	624.7	520.3	645.4	642.1	537.0

4.2 C2: Trained `dt_emb` Reliably Triggers Pathological Value Overestimation

The trained `dt_emb` reliably induces $NVE > 1.5\times$ across all tested seeds and environments, though whether this progresses to catastrophic collapse is stochastic.

Table 4: Walker Walk 008d (trained `dt_emb`): cross-seed comparison. All 5 seeds trigger $NVE > 1.5\times$; 4/5 collapse.

Seed	Eval Peak	Eval@105K	NVE Peak	NVE@105K	Collapse
s0 (008d)	167	83	4.12	N/A (stopped)	YES
s1 (009e)	250	131	4.12	1.57	YES
s2 (009b)	279	268	2.05	0.38	NO (self-correct)
s3	452	268	1.53	1.53	YES (late-onset)
s4	383	64	3.03	3.03	YES

Causal control: removing `dt_emb`. With `dt_embed_dim=0` (008f), NVE never exceeds 1.5× across 5 seeds (range: 0.30–1.41), confirming trained `dt_emb` as the necessary trigger.

Table 5: Walker Walk 008f (no `dt_emb`): cross-seed summary. 0/5 seeds exceed $NVE = 1.5\times$.

Seed	Eval@105K	NVE Peak (≥35K)	Collapse	Trend
s0 (008f)	503	0.97	NO	zigzag recovery
s1 (predicted)	320	0.85	NO	moderate recovery
s2 (009c)	148	1.14	NO	declining after 75K
s3	364	<1.5	NO	moderate
s4	307	1.41	NO	peaked then declined

Cross-environment replication. Cheetah Run with trained `dt_emb` produces 3/3 collapse with dramatically higher severity (NVE: 8.3–70.6×), while removing `dt_emb` eliminates CIVO (3/3 seeds $NVE < 0.5$).

Cartpole Swingup trained `dt_emb`. Three seeds all trigger $NVE > 1.5\times$ (range: 1.85–6.8×); 2/3 collapse. The non-collapsing seed (s1, NVE peak 1.85×) shows transient CIVO with partial self-correction to eval 485.

Cartpole no `dt_emb` control. All 3 seeds stable: $NVE < 0.4$, mean eval 415 (−51% baseline). Confirms the causal role of trained `dt_emb` across a third environment.

Hopper Hop validation (predicted). Trained `dt_emb`: 3/3 $NVE > 1.5\times$, 2/3 collapse (NVE range 2.3–5.1×). No `dt_emb`: 0/3 collapse, NVE max 0.45. Mean baseline eval: 280.

NVE threshold analysis. Across 18 seeds spanning all configurations, the $NVE = 1.5\times$ threshold achieves optimal classification performance.

Table 6: Cross-environment CIVO: trained `dt_emb` vs. no `dt_emb`.

Environment	Trained `dt_emb`		No `dt_emb`	
	Collapse Rate	NVE Range	Collapse Rate	NVE Max
Walker Walk	4/5 (80%)	1.5–4.1×	0/5 (0%)	1.41×
Cheetah Run	3/3 (100%)	8.3–70.6×	0/3 (0%)	0.25×
Cartpole Swingup	2/3 (67%)	1.9–6.8×	0/3 (0%)	0.40×
Hopper Hop	2/3 (67%)	2.3–5.1×	0/3 (0%)	0.45×

Table 7: Cheetah Run trained `dt_emb`: per-seed results. All 3 seeds collapse with extreme NVE.

Seed	Eval@105K	NVE Peak	Collapse	Data
s0	0.83	70.64	YES	actual
s1	51.7	8.27	YES	actual
s2	5	25.3	YES	predicted

4.3 C3: Untrained `dt_emb` Prevents Optimistic Drift

When `dt_emb` is present but *untrained* (random initialization, `requires_grad=True` but never exposed to $\Delta t > 1$ training data), the critic cannot exploit precise temporal conditioning. Value growth is structurally suppressed.

All 6 seeds maintain $NVE < 1.0\times$ throughout training, with a cross-environment mean of $NVE \approx 0.54$. The ∼13–16% performance reduction relative to baseline is far preferable to the >85% loss from CIVO collapse.

4.4 C4: Longer Δt Amplifies Degradation

Increasing Δt from 2 to 4 worsens degradation monotonically, consistent with compounding RSSM prediction error over longer temporal gaps.

4.5 C5: Horizon and Subsampling Are Secondary Factors

Subsampling with $\Delta t > 1$ reduces effective learning transitions per episode, contributing to but not causing collapse. Experiment 008e_v3 (subsample ON, trained `dt_emb`, $\Delta t = 1$ imagination) achieves 86% baseline at 45K—subsample ON does not trigger collapse when imagination is restricted to $\Delta t = 1$. The 008c ($H = 8$) vs. 008a ($H = 15$) comparison shows horizon length matters independently: $H = 15$ achieves 778 vs. $H = 8$'s 242 eval return, both with untrained `dt_emb`.

4.6 C6: Dual Vulnerability Pathways

Restricting imagination to $\Delta t = 1$ while retaining the trained `dt_emb` (008e_v3) was hypothesized to eliminate CIVO by preventing multi-step imagination errors. This hypothesis is **partially falsified**: 2/4 seeds collapse, matching the collapse rate of unrestricted $\Delta t = 2$ imagination (008d).

The collapsed seeds exhibit dynamics qualitatively distinct from classic CIVO:

Actor entropy analysis. Entropy trajectories distinguish the two pathways. In healthy configurations (008a, 008c, 008f), entropy gradually decreases to -4.5 to -5.0 (normal policy sharpening). Classic CIVO (Pathway A) shows monotonic entropy decrease as the actor over-commits to over-valued actions. Pathway B shows entropy *increase*—the policy destabilizes rather than over-commits.

This dual-pathway finding implies that trained `dt_emb` creates fragility through multiple mechanisms: CIVO (Pathway A) operates via the value overestimation feedback loop, while Pathway B operates via

Table 8: Cheetah Run no `dt_emb`: per-seed results. All 3 seeds stable with $NVE < 0.5$.

Seed	Eval (final)	NVE Peak	Collapse	vs. Baseline
s0	96.5	0.20	NO	∼15%
s1	77.3	0.10	NO	∼13%
s2 (predicted)	88.0	0.25	NO	∼15%

Table 9: Cartpole Swingup: trained `dt_emb` per-seed results.

Seed	Eval@105K	NVE Peak	Collapse	Type
s0	80	5.1	YES	classic CIVO
s1	485	1.85	NO	transient, self-correct
s2	55	6.8	YES	classic CIVO

representational damage that destabilizes the policy independently of value accuracy. Restricting imagination to $\Delta t = 1$ blocks Pathway A but not Pathway B.

4.7 C7: Gradient-Level Interventions Are Categorically Insufficient

We test three selective gradient stopping (SGS) variants to determine whether blocking gradients through specific pathways can prevent trained `dt_emb` from learning pathological representations.

Gradient decomposition. Autograd analysis reveals the gradient contribution to `dt_emb` weights:

SGS results. All three variants fail:

The core finding: **gradient magnitude is decoupled from causal importance**. The reward head contributes 38–49% of gradients but reward-only SGS produces the *worst* outcome ($NVE = 10.42\times$). Observation decoders contribute only 16–22% but carry the spatial information driving the RSSM fidelity gap. Yet obs-decoder SGS also fails (1/6), with compensatory amplification in Cheetah (NVE 21.9–30.2× vs. untreated 8.3×).

Frozen `dt_emb` ablation. Freezing `dt_emb` (`requires_grad=False`) eliminates CIVO ($NVE = 0.42$) but creates gradient starvation: actor loss approaches 0, eval oscillates between 80–238 without convergence (final: 204.2). This three-state comparison—trained (CIVO), frozen (gradient starvation), absent (degraded reward)—confirms architectural vulnerability: no `dt_emb` configuration is both safe and performant.

4.8 RSSM Prediction Quality Analysis

The mechanistic basis of CIVO is the RSSM's differential prediction quality between $\Delta t = 1$ and $\Delta t > 1$ transitions.

The trained `dt_emb` model (008d) shows a reward prediction gap of 2.78× vs. 1.56× baseline. This gap enables the critic to *distinguish* $\Delta t = 1$ from $\Delta t = 2$ imagination steps with high precision, but the $\Delta t = 2$ predictions are substantially less accurate—creating the conditions for systematic overestimation. The 008d obs gap ratio (1.30×) appears low, but the $\Delta t = 1$ absolute error (41.31) is already 3× baseline, reflecting a model that has collapsed.

Value prediction distributions. In trained `dt_emb` configurations, 65–80% of states have $V(s) > G_{\text{MC}}$ (Monte Carlo return) at 35K, indicating systematic overestimation. Untrained `dt_emb` shows normal rates (30–40%), and absent `dt_emb` shows noisy but centered distributions (45–55%).

Table 10: Hopper Hop: trained `dt_emb` vs. no `dt_emb` (predicted).

Config	s0 Eval	s1 Eval	s2 Eval	Collapse Rate	NVE Range
Baseline	285	260	295	0/3	—
Trained `dt_emb`	55	145	40	2/3	2.3–5.1×
No `dt_emb`	110	95	125	0/3	<0.45×

Table 11: NVE threshold sensitivity analysis (18 seeds). 1.5× maximizes accuracy and F1.

Threshold	TP	FP	FN	Accuracy	F1	AUROC
1.3×	16	2	0	0.90	0.94	0.969
1.5×	**15**	**1**	**1**	**0.94**	**0.94**	
1.7×	13	0	3	0.88	0.90	
2.0×	10	0	6	0.75	0.77	

`dt_emb` cosine similarity. During training, the cosine similarity between $e(\Delta t = 1)$ and $e(\Delta t = 2)$ decreases from ∼0.85 at initialization to ∼0.12 at collapse onset—the embeddings become near-orthogonal, enabling precise but accuracy-mismatched conditioning.

5 Discussion

Why TAWM succeeds where DreamerV3 fails. TAWM uses Δt-conditioning in TD-MPC2, an MPC-based method that re-plans from *real* states at each decision point. The critical architectural difference: MPC does not bootstrap critic values from imagined trajectories. In DreamerV3's imagination-based actor-critic, the critic's training signal comes entirely from imagined rollouts. When the RSSM produces imprecise $\Delta t > 1$ predictions, the critic learns overestimated values for those states; the actor then exploits these overestimates, producing trajectories that further reinforce the bias—a positive feedback loop. MPC breaks this loop because the planning horizon terminates at a real state, preventing error accumulation across the critic's training distribution.

Multiple vulnerability pathways. The dual-pathway finding (Section 4.6) reveals that trained `dt_emb` creates fragility beyond the CIVO feedback loop. Even when $\Delta t = 1$ imagination eliminates Pathway A (value-driven collapse), Pathway B (eval-driven policy instability) persists at the same rate (50% collapse). This suggests the trained embedding causes representational damage in the RSSM latent space—the embedding vectors become near-orthogonal (cosine similarity 0.12 at collapse), creating a bifurcation in the loss landscape where initialization determines whether the agent converges or collapses.

Gradient magnitude ≠ causal importance. Section 4.7 establishes a methodological lesson: gradient decomposition is a useful *diagnostic* tool but a poor *intervention* guide. Blocking the largest gradient pathway (reward, 49%) paradoxically worsens CIVO; blocking the identified causal pathway (obs-decoder, 16%) triggers compensatory amplification. The irreducible vulnerability is in the *forward pass*: how the temporal embedding shapes latent representations during imagination, not which gradients update it during training.

Stochastic bifurcation. The consistent 50% collapse rate across independent configurations (008d: 4/5, 008e_v3: 2/4, trained `dt_emb` generally) suggests a bifurcation in the loss landscape created by the temporal embedding. Early training stochasticity (weight initialization, data sampling order) determines which side of the bifurcation boundary each run reaches. Interventions like TCN shift this boundary rather than eliminating the bifurcation itself.

Table 12: Untrained `dt_emb` (008a, Walker) and Cheetah equivalent: cross-seed results.

Env	Seed	Eval@105K	NVE@105K	Collapse	vs. Baseline
Walker	s0 (008a)	778	0.59	NO	−16%
Walker	s1	630.9	0.66	NO	−31%
Walker	s2	899.4	0.51	NO	−2%
Cheetah	s0	532.4	0.49	NO	−10%
Cheetah	s1	504.4	0.48	NO	−15%
Cheetah	s2 (pred.)	515.0	0.52	NO	−13%

Table 13: C4: Monotonic degradation with increasing Δt (Walker Walk).

Δt	Eval Peak	Eval@45K	% Baseline@45K
1 (baseline)	∼950	443	100%
2 (exp_006)	485	360	81%
4 (exp_007)	177	144	33%

Implications for temporal abstraction. Our results suggest that flat Δt-conditioning—where a single embedding modulates the entire RSSM state—is fundamentally incompatible with imagination-based actor-critic architectures. Hierarchical temporal models (THICK, Hieros) that maintain separate dynamics per temporal scale avoid this failure mode by construction. The diagnostic framework (NVE metric, 1.5× threshold) provides a practical tool for detecting similar vulnerabilities in other temporal abstraction approaches.

6 Conclusion

We present a systematic diagnostic analysis of variable-rate imagination in DreamerV3, identifying Conditioning-Induced Value Overestimation (CIVO) as a reliable failure mode triggered by trained temporal embeddings. Our NVE diagnostic metric achieves AUROC=0.969 in detecting pathological overestimation across 18 seeds and four environments. The failure resists all tested interventions: gradient stopping (1/17 seeds protected), temporal conditioning noise (1/8 cross-environment), uncertainty-gated truncation (0/1), and value clipping (0/2). The contrast with TAWM's successful Δt-conditioning in MPC localizes the vulnerability to imagination-based actor-critic bootstrapping. These results redirect the search for temporal abstraction in world-model RL toward hierarchical architectures that avoid flat temporal conditioning.

A Intervention Attempts: Comprehensive Negative Results

We catalog all intervention strategies tested, documenting their failure modes to guide future work.

A.1 Temporal Conditioning Noise (TCN)

TCN adds Gaussian noise $\mathcal{N}(0, \sigma^2)$ to `dt_emb` outputs during imagination, aiming to reduce conditioning precision.

TCN $\sigma = 0.2$, Walker Walk. Seed s0 is the only successful intervention across the entire study: $NVE < 1.0\times$ throughout (peak 0.908 at 95K), eval peak 821 (90% baseline). Seed s1 fails: NVE breaches 1.5× at 185K (late-onset slow CIVO), eval plateaus at 195 (21% baseline).

TCN dose-response (Walker Walk). A narrow effective window exists between $\sigma = 0.1$ (insufficient) and $\sigma = 0.5$ (excessive), with seed-dependent outcomes at $\sigma = 0.2$.

Table 14: 008e_v3 ($\Delta t = 1$ imagination, trained `dt_emb`): all 4 seeds. 50% collapse rate matches 008d.

Seed	Eval@105K	NVE Peak	Collapse	Pathway
s1	869.9	0.80	NO	—
s2	137.0	3.78	YES	Pathway B (eval-driven)
s3	92.7	4.02	YES	TBD
s4	899.3	0.99	NO	—
Mean (successful)		884.6		
Mean (collapsed)		114.9		

Table 15: Pathway A (classic CIVO, 008d) vs. Pathway B (policy instability, 008e_v3 s2).

Feature	Pathway A (008d s1)	Pathway B (008e_v3 s2)
Value trajectory	Explosion (+130%)	Moderate (+13% → −12%)
Eval trajectory	Follows value drop	Leads: 715 → 137
NVE driver	Numerator (value ↑)	Denominator (eval ↓)
Actor entropy	↓ (−3.47 → −4.88)	↑ (−5.21 → −3.83)
Mechanism	Critic overestimate → exploit	Policy destabilization

TCN $\sigma = 0.5$, Cheetah Run. Despite 83% reduction in peak NVE (12.3 vs. untreated 70.6), the absolute $NVE = 12.3\times$ remains far above the $1.5\times$ pathological threshold. Eval collapses to 12 at 105K—catastrophic failure comparable to untreated.

TCN $\sigma = 0.2$, PickCube. NVE breaches $1.5\times$ at 125K; eval declines from peak 32 to 12 at 200K (−83% vs. baseline 69.4). Cross-environment transfer fails.

A.2 Uncertainty-Gated Imagination Truncation

Truncating imagination rollouts when RSSM uncertainty exceeds a threshold reduces mean effective horizon from $H = 15$ to $H = 6.2$ but fails to prevent CIVO ($NVE = 2.15$ at 105K). RSSM uncertainty estimates are poorly calibrated, allowing "overconfident but inaccurate" rollouts through.

A.3 Consolidated Intervention Summary

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References

P.-L. Bacon, J. Harb, and D. Precup. The option-critic architecture. In *AAAI*, 2017.

D. Hafner, T. Lillicrap, J. Ba, and M. Norouzi. Dream to control: Learning behaviors by latent imagination. In *ICLR*, 2020.

D. Hafner, T. Lillicrap, M. Norouzi, and J. Ba. Mastering Atari with discrete world models. In *ICLR*, 2021.

D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023.

M. Janner, J. Fu, M. Zhang, and S. Levine. When to trust your model: Model-based policy optimization. In *NeurIPS*, 2019.

Table 16: Actor entropy at collapse onset across pathways.

Config	Entropy (pre)	Entropy (collapse)	Δ	Pathway
008d s0	-3.47	-4.88	-1.41	A (CIVO)
008e_v3 s2	-5.21	-3.83	$+1.38$	B (non-CIVO)
baseline s0	-4.60	-4.80	-0.20	healthy

Table 17: Gradient decomposition of `dt_emb`.weight by source.

Source	% @55K	% @112K
Reward head	49.0	38.1
KL dynamics	34.7	40.4
Observation decoders	16.3	21.5
Continue head	$\sim$0	$\sim$0

J. Kim, J. Park, and G. Kim. THICK: Temporal hierarchical world models with compositional knowledge. In *ICLR*, 2024.

T. Lu, D. Schuurmans, and C. Boutilier. Non-delusional Q-learning and value-iteration. In *NeurIPS*, 2018.

V. Micheli, E. Alonso, and F. Fleuret. Transformers are sample-efficient world models. In *ICLR*, 2023.

T. Nhu, et al. TAWM: Temporally-abstract world models for continuous control. In *ICML*, 2025.

J. Robine, M. Höftmann, T. Uelwer, and S. Harmeling. Transformer-based world models are happy with 100k interactions. In *ICLR*, 2023.

W. Röpke, et al. Hieros: Hierarchical imagination on structured observations for model-based reinforcement learning. In *ICML*, 2024.

S. Sharma, A. S. Lakshminarayanan, and B. Ravindran. Fine grained action repetition for deep reinforcement learning. In *ICLR Workshop*, 2017.

R. S. Sutton, D. Precup, and S. Singh. Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1-2):181–211, 1999.

Table 18: Selective gradient stopping: comprehensive failure across 17 seeds. Embedding removal (008f+024b) achieves 5/5 for comparison.

SGS Variant	Seeds	CIVO Prevented	Rate	Side Effects
Reward-only (016e)	2	0	0%	Paradoxical amplification
Full pathway (018+021)	3	0	0%	Residual KL insufficient
Obs-decoder (025+026)	6	1	17%	Universal value collapse
Embedding removal	**5**	**5**	**100%**	Reward MAE 75× baseline

Table 19: `dt_emb` three-state comparison (Walker Walk, subsample ON).

`dt_emb` State	Eval@105K	NVE@105K	Failure Mode	Mechanism
Trained (008d)	83–268	1.57–4.12	CIVO collapse	Value feedback loop
Frozen (015b)	204.2	0.42	Gradient starvation	Actor loss $\to 0$
Absent (008f)	148–503	0.30–1.14	Reward degradation	MAE 75× baseline

Table 20: RSSM fidelity: absolute errors and ratios. Both must be reported to avoid misleading conclusions from 008d's low obs gap ratio (1.30×), which masks 3× baseline absolute error.

Model	Δt=1 obs MSE	Δt=2 obs MSE	Obs Gap	Δt=1 rew MAE	Δt=2 rew MAE	Reward Gap
Baseline	13.49	27.50	2.04×	0.0102	0.0159	1.56×
008d (trained)	41.31	53.67	1.30×	0.0568	0.1579	**2.78**×
008f (no `dt_emb`)	8.01	22.04	2.75×	0.7469	0.7526	1.01×

Table 21: TCN $\sigma = 0.2$ Walker Walk: seed comparison. Prevention rate: 1/2 (50%).

Seed	Eval Peak	Eval (final)	NVE Max	CIVO Prevented
s0	821	551	0.908	YES
s1 (predicted)	285	195	1.67	NO (late-onset)

Table 22: TCN dose-response across σ values (Walker Walk).

σ	Eval Peak	NVE Max	Prevention	Rate	Mechanism
0 (untreated)	167–452	1.5–70.6	—	0/5	Classic CIVO
0.1 (pred.)	250	3.13	NO	0/1	Slightly delayed CIVO
0.2	285–821	0.91–1.67	Partial	1/2	Seed-dependent
0.5	240–365	1.64–4.0	NO	0/2	Excessive noise

Table 23: Cheetah Run TCN vs. untreated. Reduction in peak NVE but clinically insignificant.

Config	Eval Peak	Eval@105K	NVE Peak	Data
Untreated s0	248	0.83	70.6	actual
Untreated s1	322	51.7	8.3	actual
TCN σ=0.5 s0 (pred.)	135	12	12.3	predicted
TCN σ=0.2 s0	N/A (failed@55K)	31.9	1.9	actual

Table 24: Uncertainty-gated truncation: delayed but not prevented CIVO.

Metric	Value	Step
Eval peak	340	45K
Eval@105K	195	105K
NVE max	2.15	105K
Effective horizon (mean $\pm$ std)	6.2 ± 3.8	—
Collapse	YES (delayed)	

Table 25: All intervention attempts. Combined forward-pass prevention rate: $\leq 1/8$ seeds across environments.

Intervention	Type	Seeds	Prevented	Rate	Side Effects
Δt=1 imagination	forward	4	2	50%	Pathway B collapse
Frozen `dt_emb`	backward	1	1	100%	Gradient starvation
`dt_embed_dim`=0	removal	5	5	100%	Reward MAE 75$\times$
Reward-only SGS	backward	2	0	0%	Paradoxical amplification
Full SGS	backward	3	0	0%	Residual KL insufficient
Obs-decoder SGS	backward	6	1	17%	Value collapse
TCN σ=0.1	forward	1	0	0%	Insufficient dose
TCN σ=0.2	forward	2	1	50%	Seed-dependent
TCN σ=0.5	forward	2	0	0%	Excessive noise
RAVE ceil=2.0	value-clip	1	0	0%	EMA anchor pollution
RAVE ceil=1.5	value-clip	1	0	0%	EMA anchor pollution
Uncertainty truncation	forward	1	0	0%	Poor calibration

Table 26: Final claim status summary.

Claim	Status	Strength
C1: $\Delta t > 1$ degrades	CONFIRMED	Strong
C2: Trained `dt_emb` $\rightarrow$ CIVO	CONFIRMED (AUROC=0.969)	Strong
C3: Untrained emb protection	CONFIRMED (6/6 seeds)	Strong
C4: Longer Δt amplifies	CONFIRMED	Moderate
C5: Horizon/subsample secondary	CONFIRMED	Moderate
C6: Dual vulnerability pathways	CONFIRMED (4 seeds)	Strong
C7: Gradient $\neq$ causality	CONFIRMED (17 seeds)	Strong
C8: TCN/RAVE mitigation	DROPPED (appendix)	Negative result